

LESSON 9-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 1 OF 2**Substitute and Solve**

Maryland Standard 6.EE.C.9 • Learning target – I can substitute a value into an equation to find the matching output.

REMEMBER

- To find the output, **substitute** the input for x .
- Then use **order of operations**: multiply first, then add or subtract.

WORKED EXAMPLEUse $y = 6x + 4$ to find y when $x = 5$.**Step 1** Substitute: $y = 6 \times 5 + 4$ **Step 2** Multiply, then add: $30 + 4 = 34$ **Answer $y = 34$, so (5, 34)****WORD BANK****substitute** – put the number in place of the letter**output** – the answer you get (y)**order of operations** – multiply first, then add or subtract**PRACTICE***show your work • label units*

- 1.**
- Substitute to find
- y
- .

Step 1 Put the number in place of x .**Step 2** Multiply first. Then add or subtract.

a. $y = 4x + 7$, $x = 3$ _____

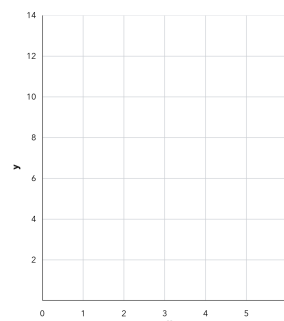
b. $y = 9x$, $x = 8$ _____

c. $y = 3x - 2$, $x = 10$ _____

d. $y = x + 15$, $x = 27$ _____

- 2.**
- Complete the table for
- $y = 2x + 3$
- . Then plot the points and connect them.

x	0	1	2	4	5
y					



- 3. Water Tank.**
- A tank has 20 gallons. It fills at 15 gallons each minute, so
- $g = 15m + 20$
- .

Step 1 Put 6 in place of m : 15×6 .**Step 2** Then add the 20 gallons.**a.** Gallons after 6 minutes: _____ gal**b.** Gallons after 10 minutes: _____ gal

- 4. Bike Rental.** A bike rental costs $c = 7h + 5$ dollars for h hours. Find the cost for **3** hours.
Then find the cost for **8** hours.

Step 1 Multiply 7 by the hours.

Step 2 Then add the \$5.

Costs _____

- 5. On the Line?** Substitute x to decide whether each point is on the graph of $y = 5x + 2$.

a. $(4, 22)$: $5 \times 4 + 2 = \underline{\hspace{2cm}}$ On the line? ☐ Yes ☐ No

b. $(6, 30)$: $5 \times 6 + 2 = \underline{\hspace{2cm}}$ On the line? ☐ Yes ☐ No

- 6. Find the Error.** For $y = 4x + 3$ with $x = 5$, Dev added first: $4 + 3 = 7$, and $7 \times 5 = 35$.
Explain his mistake and find y .

Frame Dev's mistake is _____. He should multiply first: $y = \underline{\hspace{2cm}}$.

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____

AT HOME Pick a rule like $y = 2x + 1$. Ask someone at home for an input and find the output together.

LESSON 9-4 • HOMEWORK PACKET • LEVEL 1 • SHEET 2 OF 2**Working Backward to a Goal**

Maryland Standard 6.EE.C.9 • Learning target – I can work backward from a goal to find the input an equation needs.

REMEMBER

- Know the goal (y)? Substitute it, then solve for x .
- Undo in **reverse order**: subtract the fee first, then divide by the rate.
- In a counting problem, a decimal answer may need to be **rounded up**.

WORKED EXAMPLE

A rental costs $y = 4x + 10$ for x hours. How many hours cost \$46?

Step 1 Subtract 10: $46 - 10 = 36$

Step 2 Divide by 4: $36 \div 4 = 9$

Answer 9 hours

WORD BANK

work backward – start with the goal and undo the steps

inverse operation – an action that undoes another

round up – go to the next whole number

goal – the number you want to reach

PRACTICE*show your work • label units*

1. Solve for x .

Step 1 Undo $+$ or $-$ first.

Step 2 Then divide by the number next to x .

a. $7x = 63$ _____

b. $8x = 104$ _____

c. $3x + 5 = 32$ _____

d. $6x + 12 = 60$ _____

2. **Trip Fund.** A class has \$45 and earns \$15 a week: $y = 15x + 45$. How many weeks until it has \$210?

Step 1 Subtract the \$45 the class already has.

Step 2 Divide what is left by \$15 a week.

Answer _____ weeks

3. **Gift Fund.** A club has \$30 in its gift fund. Members add \$12 each week, so $y = 12x + 30$. How many weeks until the fund has \$150?

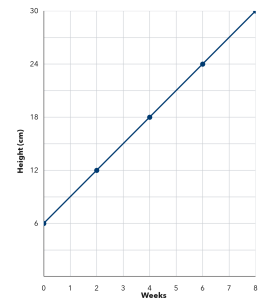
Step 1 Subtract the \$30 already in the fund.

Step 2 Divide what is left by \$12.

Answer _____ weeks

4. **Plant Growth.** A plant is 6 cm tall and grows 3 cm a week: $h = 3w + 6$.

- a. Use the graph. When is it 18 cm? _____
b. Use the equation. When is it 33 cm? _____



5. **Round Up.** A box holds 12 cookies. Ana needs 150 cookies, so she says 12.5 boxes. How many boxes should she buy? Explain.

Frame Ana should buy ____ boxes because ____.

TEST PRACTICE

choose the best answer

6. A rental costs $y = 8x + 12$ dollars for x hours. Max has \$60. How many hours can he rent?
A 5 hours **B** 6 hours **C** 7.5 hours **D** 9 hours
7. Mia solved $5x + 20 = 85$ by dividing 85 by 5 first. What is the correct value of x ?
A $x = 17$ **B** $x = 21$ **C** $x = 13$ **D** $x = 65$

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____